

Somak Laha

Ph.D. Candidate, Statistics · Harvard University

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Education

Harvard University

Ph.D. in Statistics | Advisor: Prof. Morgane Austern

Cambridge, MA

Sept 2023 – Present

Indian Statistical Institute

M.Stat. in Statistics, specialization in Applied Statistics

Kolkata, India

Sept 2020 – June 2022

Indian Statistical Institute

B.Stat. (Honours) in Statistics

Kolkata, India

July 2017 – Sept 2020

Industry Experience

Capital One Datalabs

Data Science Analyst (Full-time)

Bangalore, India

July 2022 – Aug 2023

- Built an ensemble statistical/ML model predicting used-car trade-in values for Capital One's AutoNavigator platform.
- Developed no-reference image quality assessment to gate car-image preprocessing, evaluating CLIP, SAM, MANIQA and YOLO.

Capital One Datalabs

Data Analyst Intern

Bangalore, India

June – July 2021

- Improved the design and interpretation of production A/B tests; implemented a Bayesian A/B testing widget in Python for analyst use.

Preprints

The Zero Pattern of a Design Matrix Drives Multiple Descent in Over-parameterized Regression

[arXiv:2607.24041](https://arxiv.org/abs/2607.24041)

July 2026

K. H. Huang, H. Ye, S. Laha, M. Austern

- Studied multiple peaks in test error in over-parameterized linear regression, showing that they can arise from heterogeneity and dependence in the design rather than from changes in model architecture.
- Derived deterministic equivalents for the prediction bias and variance under heterogeneous or dependent Gaussian designs with possibly singular covariance matrices, in a vanishing-ridge regime.
- Showed that the zero pattern of the induced variance profile determines where additional test-error peaks can occur, and validated the theory through synthetic experiments, data augmentation, and real gene-expression data pooled with GAN-generated samples.

Graph Attention Network for Node Regression on Random Geometric Graphs with Erdős–Rényi Contamination

[arXiv:2601.23239](https://arxiv.org/abs/2601.23239)

Jan 2026

S. Laha, S. Liu, M. Austern

- Graph attention networks cope well with noisy features and spurious edges in practice, but there is little theory showing that attention actually beats a plain graph neural network.
- We study node regression where the response depends on latent covariates but the observed covariates are noise-contaminated, and the graph is a random geometric graph on the latent covariates polluted by independent Erdős–Rényi edges. The geometric edges carry information about the latent positions; the contaminating edges carry none.
- Our designed attention layer scores neighbours so that the geometric ones dominate, builds denoised proxies for the latent covariates, and regresses on those. Cross-fitting keeps the proxy construction from biasing the regression that uses it.
- The proxies beat ordinary least squares on the noisy covariates for estimating the coefficient, and beat a vanilla GCN for predicting at an unlabelled node, both under mild growth conditions.

Research Experience and Academic Projects

Gaussian Coupling Bounds with Explicit Constants via Stein Kernels

Cambridge, MA

With Prof. Morgane Austern, Department of Statistics, Harvard University

July 2026 – Present

- How close is a normalized sum to the Gaussian it is quantile-coupled with? Known bounds either miss the right rate in n , decay too slowly in the tail, or leave the constants unspecified.
- Using Stein kernels and the Ornstein–Uhlenbeck interpolation, we get non-asymptotic tail bounds on the coupling error with explicit constants and the same tail shape as its limiting distribution.

- When the third moment vanishes, second-order Stein kernels give a tighter tail approximation, at rate $1/n$ up to a log factor.

Symmetry Breaking and Collective Bias in Populations of LLM Agents

Cambridge, MA

With Prof. Subhabrata Sen, Department of Statistics, Harvard University

June 2026 – Present

- Ask a single LLM which letter of the alphabet it prefers and it looks unbiased. Put a population of them in a naming game and one letter wins the convention far more often than chance. The suspicion is that a tiny bias, invisible in any one agent, is amplified by repeated communication.
- We model the population with Quantized Simplex Gossip, in which agents hold internal belief states and learn from one another's sampled outputs, so one agent's arbitrary draw becomes the next agent's evidence.
- We start the agents just barely off symmetry to investigate how small a bias can be amplified into the convention.

Depreferential Attachment Model through Preferential Attachment

Kolkata, India

Advisor: Prof. Antar Bandyopadhyay, Statistics and Mathematics Unit, ISI

Jan 2022 – Present

- Preferential attachment lets the largest vertices keep growing fastest, which is what produces power-law degrees. Many real systems push back against that.
- The model damps the leaders without inverting the attachment rule. Each step samples a set of old vertices in a degree-biased way, forbids them, then runs an ordinary preferential attachment step on whoever is left.
- Dissertation work: bounded the maximal degree under hard-core tabooing, and proved upper and lower bounds on the asymptotic order of a fixed vertex's degree and of the degree distribution under the soft-core scheme, with simulations alongside.
- Ongoing work studies fixed vertex degree and degree distribution asymptotics in sharper ways under varying taboo rate. It proves central limit theorems for fixed vertex degrees using martingale CLT techniques.
- [Dissertation presentation](#)

Edge-preserving Regression Surface Fitting

Kolkata, India

Advisor: Prof. Partha Sarathi Mukherjee, ISRU, ISI | Group: Riddhiman Saha, Souhardya Ray

Aug 2020

- Proposed a two-stage procedure that recovers discontinuous regression surfaces from noisy data. Neighborhood-based clustering finds the jumps, then local polynomial kernel smoothing fits around them.
- [Project report & Presentation](#)

Linear Models Projects: Shape Analysis and Satellite Image Stitching

Kolkata, India

Advisor: Prof. Arnab Chakraborty, Applied Statistics Unit, ISI

Aug – Nov 2019

- Fitted a Fourier descriptor model for convex shapes using linear mixed effects models, and wrote the fitting routine and an accompanying validity test in R.
- Estimated landmark positions from overlapping satellite screenshots via linear models and stitched them into a single vector map (R).

Teaching

Fall 2026	Teaching Fellow , STAT 211: Statistical Inference I (graduate)	Harvard University
Spring 2025	Teaching Fellow , STAT 212: Probability II (graduate)	Harvard University
Fall 2024	Teaching Fellow , STAT 120: Introduction to Bayesian Inference and Applications	Harvard University
Led weekly sections, held office hours, and graded problem sets and exams.		

Talks and Presentations

Graph Attention Network for Node Regression on Random Geometric Graphs with Erdős–Rényi Contamination

Joint Statistical Meetings (JSM) 2026

Boston, MA · Aug 2026

2025 IMS International Conference on Statistics and Data Science (ICSIDS)

Seville, Spain · Dec 2025

Awards and Scholarships

2020	Academic Award , Prize for outstanding academic performance, 5th semester	Indian Statistical Institute
2017	IIT-JEE Advanced , All India Rank 676	India
2016	KVPY Fellowship , All India Rank 689, SX stream	DST, Govt. of India

Skills

Computing Python, R, C, SQL (Snowflake)